

2. Explain the difference between each of the three views

The three choropleth maps differ in how the data is classified into ranges, which affects how patterns appear across Ohio counties.

Quantile classification divides the data so that each class contains the same number of counties. This means each color category has an equal number of observations, but the value ranges within each class may be very different. Quantile maps are useful for comparing counties relative to each other, but they can sometimes exaggerate differences when values are very close.

Equal Interval classification divides the data into classes with equal numeric ranges. For example, if values range from 0–100 and there are five classes, each class represents a range of 20 units. This method makes the legend easy to understand, but it can cause most counties to fall into the same category if the data is unevenly distributed.

Jenks Natural Breaks classification groups data based on natural clusters within the dataset. The algorithm minimizes variance within each class and maximizes differences between classes. This method often provides the most accurate representation of real patterns in the data because it reflects where natural groupings occur.

Quantile filtered view is similar to the quantile classification but removes or filters out certain values allowing the map to focus more clearly on the remaining distribution.

3. Explain how this form of GIS mapping can be used to improve public/population health

Choropleth maps are useful in public and population health because they allow researchers and policymakers to visualize how health outcomes or risk factors vary geographically. By mapping variables such as disease rates, access to healthcare, socioeconomic conditions, or environmental exposures, public health officials can quickly identify areas with higher risks or disparities. This information can guide resource allocation, target interventions, and support data-driven decision making to improve health outcomes in specific communities.

4. This assignment took me 1 hour to complete.

5. I would rate this a 2 out of 5 on difficulty.

Extra Credit:

Interpret the view. Can you make any inferences from the plot and is it useful? Does changing the categorization technique influence the interpretation?

The population choropleth map shows how population is distributed across ZIP codes in Ohio. From the map, it can be inferred that areas with higher population densities tend to be

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concentrated around major cities such as Cleveland, Columbus, Cincinnati, and other metropolitan areas. Rural areas tend to show much lower population levels. This type of visualization is useful because it helps identify where large numbers of people live, which can influence how resources such as healthcare services, vaccination sites, or testing locations should be distributed.

Changing the categorization technique can influence how the data is interpreted. Different classification methods group the population values differently, which may highlight or hide certain patterns. For example, Equal Interval may place many ZIP codes into the same category if population values are unevenly distributed, while Jenks Natural Breaks tends to better highlight natural clusters in the data. As a result, the choice of classification method can affect how clearly population differences appear on the map and may influence the conclusions drawn from the visualization.